

UCT601: COMPUTER NETWORKS

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3	0	2	4.0

Course Objectives: To build the fundamental concepts of computer network, networking devices and various networking protocols.

Introduction: Computer networks and distributed systems, Classifications of computer networks, Preliminaries of layered network structures.

Data communication Components: Representation of data and its flow, Various Connection Topology, Protocols and Standards, OSI model, Transmission Media.

LAN: Wired LAN, Wireless LAN, Virtual LAN.

Techniques for Bandwidth utilization: Multiplexing - Frequency division, Time division and Wave division, Concepts on spread spectrum.

Data Link Layer and Medium Access Sub Layer: Fundamentals of Error Detection and Error Correction, Block coding, Hamming Distance, CRC; Flow Control and Error control protocols - Stop and Wait, Go-back-N ARQ, Selective Repeat ARQ, Sliding Window, Piggybacking, Random Access, Multiple access protocols - Pure ALOHA, Slotted ALOHA, CSMA/CD, CDMA/CA.

Network Layer: Switching, Logical addressing – IPV4, IPV6; Address mapping – ARP, RARP, BOOTP and DHCP–Delivery, Forwarding and Unicast Routing protocols.

Transport Layer: Process to Process Communication, User Datagram Protocol (UDP), Transmission Control Protocol (TCP), SCTP Congestion Control; Quality of Service (QoS), QoS improving techniques - Leaky Bucket and Token Bucket algorithms.

Application Layer: DNS, DDNS, TELNET, EMAIL, FTP, WWW, HTTP, SNMP, Bluetooth, Firewalls.

Network Security: Electronic mail, directory services and network management, Basic concepts of Cryptography.

Laboratory Work:

Network commands; ping, ipconfig/ifconfig, getmac, arp, hostname, tracert, netstat, nslookup, path ping etc. Working of various network layer protocols in an open source simulator.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

Upon completion of this course, students will be able to:

1. Describe the basis and structure of an abstract layered protocol model.
2. Identify the different types of network devices and their functions within a network.

3. Comprehend and building the skills of subnetting and routing mechanisms.
4. Describe, analyse and evaluate a number of data link, network, and transport layer protocols.

Text Books:

1. Forouzan A. B., Data communication and Networking, McGraw Hill (2012) 5th ed.
2. Tanenbaum S. A. and Wetherall J. D., Computer Networks, Prentice Hall (2013) 5th ed.

Reference Books:

1. Kurose J. and Ross K., Computer Networking: A Top Down Approach, Pearson (2017) 7th ed.
2. Stallings W., Computer Networking with Internet Protocols and Technology, Pearson (2004).